

Analyse Your Data with Pandas

Here's an introduction to Pandas, an open source software library that's written in Python for data manipulation and analysis. Pandas facilitates the manipulation of numerical tables and the time series.



In recent times, it has been proven again and again that data has become an increasingly important resource. Now, with the Internet boom, large volumes of data are being generated every second. To stay ahead of the competition, companies need efficient ways of analysing this data, which can be represented as a matrix, using Python's mathematical package, NumPy. The problem with NumPy is that it doesn't have sufficient data analysis tools built into it. This is where Pandas comes in. It is a data analysis package, which is built to integrate with NumPy arrays. Pandas has a lot of functionality, but we will cover only a small portion of it in this article.

Getting started

Installing Pandas is a one-step process if you use Pip. Run the following command to install Pandas.

```
sudo pip install pandas
```

If you face any difficulties, visit <http://pandas.pydata.org/pandas-docs/stable/install.html>. You can now try importing Pandas into your Python environment by issuing the following command:

```
import pandas
```

In this tutorial, we will be using data from Weather Underground. The dataset for this article can be downloaded from http://www.synesthesiam.com/assets/weather_year.csv and can be imported into Pandas using:

```
data = pandas.read_csv('weather_year.csv')
```

The `read_csv` function creates a dataframe. A dataframe is a tabular representation of the data read. You can get a summary of the dataset by printing the object. The output of the print is as follows: